RCP2027
User Manual

PROMIR-BUD PASSION LINE

Using the desktop application and terminal after installation. SD card configuration, offline and online operation, employee management and door-release permissions.

Who is this manual for?
Administrators managing the SD card and employee records, and employees recording attendance with RFID cards. This manual assumes the firmware is installed and the desktop application is available.

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Version and interface labels

English edition of the updated manual, based on the desktop application and firmware in the RCP2027 project reviewed on 11 October 2026. Hardware and Google integration procedures are derived from the implementation; this manual is not a hardware test report.

Four interface languages: the desktop application and terminal support Polish, English, German and Spanish. This manual uses English names for application controls. Select English in the application interface language options and separately set the terminal language. Control wording may vary slightly between application versions.

Wallpaper is supported: prepare the image in the desktop application, export it to the SD card and enable it in the terminal settings. See page 10.

1. Application and everyday use

RCP2027 records employee clock-in and clock-out events using RFID cards. The terminal includes a touchscreen, clock, SD card, door-release relay support and an optional camera. The desktop application prepares the configuration and manages data tables.

Desktop tab	Purpose
Configuration and SD card	Choose the SD folder, mode and language; configure Wi-Fi, API, spreadsheets, time offset and photos. Save settings or prepare the card.
Wallpaper	Prepare a 480 × 320 image, preview interface areas and export wallpaper files to SD.
Employees	Add, edit or remove records; activate cards and assign door-release permissions.
Attendance	View and manually correct clock-in and clock-out records.
Logs	View diagnostic messages. This table is read-only.

Clocking in and out

1. Present an active employee card to the reader.

2. A card without relay permission automatically records the next event: clock-out after clock-in, and clock-in after clock-out. If there is no history, the first event is clock-in.

3. For a card with relay permission, actions depend on the last attendance event: an employee outside can select **Clock in**; after clock-in, the options are **Clock out** or **Unlock door**. During a Wi-Fi outage, select clock-in or clock-out manually; door release is unavailable.

4. Check the confirmation screen for **Good morning** or **Goodbye** and the employee name. It is displayed for approximately 3 seconds. Distinguish a confirmed record from a queued event or an unknown result; an event awaiting synchronization is not yet a confirmed Google record.

> **Door release and attendance: Unlock door** activates the relay for 3 seconds and creates a log entry. It does not create a clock-in or clock-out record. Clock in and Clock out do not themselves activate the relay.

Offline and online data

Offline mode uses employee and attendance files on SD. Online mode reads employees from Google and records events through Apps Script. Logs are stored on SD in both modes and additionally sent to Google in online mode. Photos remain on SD.

1. Using the desktop application

Recommended workflow

Open RCP2027 - Time Attendance Recorder and select **Configuration and SD card**. The application manages local data through the SD card and online data through the Google API; it does not use a USB cable connection to the terminal.

1. Click **Select SD card** and choose the root folder. If config.json exists, its settings are loaded. Offline employee data is also loaded; attendance and log files must be selected separately.

2. Check **Operating mode**. It determines whether Refresh and Save / send changes use SD or Google. Changing this field alone does not change the terminal mode: save the configuration and restart the terminal with that card.

3. Complete and save the configuration form. Then open the relevant data tab, refresh or select its file, edit the table and save those changes separately.

4. Prepare wallpaper in **Wallpaper**. Its export folder is selected separately; selecting SD in the configuration tab does not automatically set the wallpaper output folder.

Configuration buttons

Button	Action
Save configuration	Writes the form to config.json. Does not save employee or attendance tables.
Prepare SD card	Saves configuration and creates missing CSV files and the photos folder. Preserves existing CSV records.
Check links	Checks Google URL formats; does not test the connection.
Test API	Tests connection, authorization, spreadsheets and headers.
Generate token	Generates a token in the form. Copy the same token to Apps Script properties.
Help	Opens Instructions and Apps Script code. Copy code copies the supplied script.

Editing tables and waiting for requests

Edit employee and attendance cells directly. **Add record** inserts a blank row. **Remove selected** removes every row containing a selected cell. Only **Save / send changes** commits edits. Refreshing replaces unsaved table contents.

When the application displays Connecting to Google Apps Script, wait for the result. API buttons are temporarily disabled while the window remains responsive. Check the completion message before the next operation.

2. Setting up offline mode

Offline mode works without Internet access. The SD card supplies employee records and stores attendance events.

1. Turn off the terminal before removing its SD card. Connect the card to the computer and back up existing data.

2. In **Configuration and SD card**, click **Select SD card**. Select the card root, for example E:\, rather than a subfolder.

3. Under **Operating mode**, select **Offline - SD card**. Choose the terminal language and time-zone offset in minutes. Google and Wi-Fi fields are not required for offline operation.

4. Choose whether to enable photo capture at clock-in/out. It requires a working camera and free SD space.

5. Click **Prepare SD card**. If asked to replace config.json, confirm only when the form contains the intended settings.

6. Add employees as described in section 4, or start with an empty employee list and enroll cards at the terminal.

7. Safely eject the card, insert it into the powered-off terminal and switch the terminal on. Check **OFFLINE**, the date and time, then make a test attendance record with an active card.

SD item	Purpose
config.json	Terminal settings. Preparing SD writes the current configuration form.
employees.csv	Employee records and card permissions.
attendance.csv	Clock-in and clock-out events.
logs.csv	System diagnostics.
photos/	JPEG photos when photo capture is enabled.

Selecting an SD folder automatically initializes missing or empty CSV files with headers. Preparing SD preserves existing CSV records and creates photos/. **Save configuration** saves config.json only; save employee table changes separately.

Changing mode at the terminal: Settings → PIN → Operating mode toggles offline/online and saves the configuration to SD. Offline mode loads employees.csv. This file may contain the cached list from a previous online synchronization; check that the local list is complete before switching to offline.

3. Online: preparing the connection

Online mode requires Wi-Fi and a working Google Apps Script API. SD is still required for configuration, local logs, photos and the attendance queue.

If Google integration is already configured

Obtain the Wi-Fi network name and password, Apps Script deployment URL ending in /exec, API token, and links to the employee, attendance and log spreadsheets from the administrator. Continue with SD configuration on page 6.

Creating the integration

1. Create three Google spreadsheet documents. In the first tab of each document, put the appropriate headers in the first row, one header per column, in the exact order shown below.

2. Create an Apps Script project and paste the supplied **google_apps_script/Code.gs** code. The desktop **Help** button provides instructions and code to copy.

3. Click **Generate token** in the desktop application. Set the following Apps Script project properties. The script token and application token must match.

4. Deploy as a web app that executes as the owner and is accessible to everyone. API operations are protected by the token. Copy the /exec deployment URL. After changing Code.gs, publish an updated deployment version.

Script property	Value
RCP_API_TOKEN	Token from the application, at least 32 characters.
EMPLOYEES_SHEET_ID	Employee spreadsheet document ID.
ATTENDANCE_SHEET_ID	Attendance spreadsheet document ID.
LOGS_SHEET_ID	Log spreadsheet document ID.

The document ID is the URL segment between /spreadsheets/d/ and the next slash. A spreadsheet link identifies the document; writes are performed through Apps Script.

Exact first-row headers

Employees:
employee_id, card_uid, first_name, last_name, active, relay_access

Attendance:
event_id, timestamp, employee_id, card_uid, event_type, photo_file, source, notes

Logs:
event_id, timestamp, level, code, message

3. Online: SD configuration

1. Turn off the terminal, connect its SD card to the computer and back up the data. Click **Select SD card** and select the root folder.

2. Choose **Online - Google Sheets**. Set the language, Wi-Fi network name and password, and time-zone offset. Enter the Apps Script /exec URL.

3. Paste the employee, attendance and log spreadsheet links into their respective fields. Enter the token matching RCP_API_TOKEN in the script properties.

4. Click **Check links**. This checks URL format, not service connectivity.

5. Click **Test API**. A successful result reads: The API and configured Google spreadsheets are accessible. The test checks authorization, headers and spreadsheet IDs.

6. Click **Prepare SD card**, or **Save configuration** on an already prepared card. Preparing SD may display local employees in the table; open **Employees** and click **Refresh** before any online editing.

7. Safely eject the card, insert it into the powered-off terminal and restart. Check ONLINE, the clock and network/API messages. Make a test record and verify it in Google attendance data.

Handling network outages:
The terminal remains ONLINE. Without Wi-Fi it recognizes cards from the cached employee list and asks for manual clock-in or clock-out selection. Events are saved in pending_online_attendance.csv on SD and uploaded in order after connectivity returns. A queued event is removed only after server confirmation. Online events are not written to the local attendance.csv file.

If a request result is unknown, the terminal retains the event ID, blocks further attendance records and checks the event status. Wait for resolution and synchronization. Do not delete the queue or repeat the attendance action assuming the first failed. Outage recording requires working SD and RTC and an employee card recognized from the cached list.

Employee synchronization

The terminal loads employees at startup and refreshes approximately every 5 minutes. Changes made in the desktop application or spreadsheet may therefore be delayed. Restarting can initiate a new load; verify that synchronization succeeded.

Configuration storage

The Wi-Fi password and API token are stored as plain text in config.json on SD. Protect the card and backups. If the card is lost, replace the token in both Apps Script and the device configuration.

The desktop API test checks the computer connection. It does not replace a Wi-Fi and attendance test on the terminal.

4. Employees and door access: offline

Adding employees in the desktop application

1. Select the SD card, choose Offline and open **Employees**. Click **Refresh** to load employees.csv. Refreshing replaces unsaved edits.

2. Click **Add record** and complete all columns below. Copy the card UID exactly. Enrolling at the terminal is easiest when the UID is unknown, because the terminal reads it automatically.

3. Click **Save / send changes** and wait for file-save confirmation. Insert the SD card into the powered-off terminal and restart.

Column	What to enter
employee_id	Unique ID, for example P001. Keep it consistent with employee history.
card_uid	Unique RFID UID, for example A1B2C3D4 (fictional example).
first_name / last_name	Employee first name and surname.
active	true = active; false = card is not authorized.
relay_access	true = permission to use door release; false = no permission.

Adding employees at the terminal

Present an unknown card, choose **Yes** and enter the settings PIN. Enter the ID, first name and surname, select **Relay access: YES/NO** and tap **Save**. The employee is active and saved to employees.csv. Alternatively use Settings → PIN → Employees → Add employee, then scan the new card and enter the PIN.

Granting or removing door permission

In the desktop application, set relay_access to true or false and save. At the terminal, open Settings → PIN → Employees and tap **Relay** beside the person: green indicates permission and red indicates no permission. Terminal changes are saved immediately.

Removing and reactivating employees

Terminal **Remove** sets active=false and hides the employee from the active list, preserving attendance history. In the application, set active=false for the same result. To reactivate, set active=true in the existing record and save.

Desktop **Remove selected** deletes entire selected records when the table is saved. It does not delete attendance events, but deactivation is recommended to keep employee details linked to history. Do not enroll a deactivated card again as a new employee with the same UID.

5. Employees and door access: online

Adding and editing through the desktop application

1. Select Online in configuration and check the Apps Script URL and token. Open **Employees** and click **Refresh**. Load current data before each editing session.

2. Click **Add record** or edit an existing row. Use the columns on page 7. IDs and UIDs must be unique, including those belonging to inactive employees.

3. Set relay_access=true to grant door permission or false to remove it. Set active=false to deactivate a card; active=true reactivates it.

4. Click **Save / send changes**. The application asks to replace the entire spreadsheet table with the current table. Check that all records are present, then confirm. Inserting a row on screen alone does not save it.

5. Wait for upload confirmation. Allow approximately 5 minutes for the terminal refresh, or restart it and verify synchronization.

> **If saving is rejected because the data changed:** the spreadsheet was modified after you loaded it. Note your intended edits, click Refresh again, apply them to the latest table and save. Refreshing discards unsaved edits in the view.

Adding employees at the terminal

Present an unknown card → Yes → PIN → ID, name and relay permission → Save. Alternatively use Settings → PIN → Employees → Add employee. The terminal appends the employee to Google through the API. Enrollment requires a working connection; it is not an offline employee-file edit.

Changing permissions or removing employees

In Settings → PIN → Employees, **Relay** toggles door permission and **Remove** deactivates the employee through the API. The change is saved after a successful request. Reactivate an inactive employee by editing their existing record in the desktop application.

Deleting an entire employee record

Select the record, click **Remove selected**, then **Save / send changes** and confirm table replacement. Deleting an employee does not remove attendance events. Deactivation better preserves the identity associated with historical events.

Editing local employees.csv does not update the online employee list. Online mode uses Google data. Restrict direct spreadsheet editing to authorized administrators.

6. Additional settings

Opening settings and changing the PIN

Tap **SETTINGS** on the home screen and enter the four-digit PIN. The default is **0000**; change it before normal use. Five of eleven settings are visible at once. Tap the right-hand scrollbar to reveal more items, including Date and time, Change PIN, Calibrate touch, Wi-Fi and Wallpaper.

In **Change PIN**, enter and confirm a new PIN, then enter it again. It is stored in terminal memory. Use **Back** to return.

Wallpaper and display colours

Prepare your wallpaper in the desktop application, export it to SD, restart the terminal and enable Wallpaper in settings. The full procedure is on page 10.

Color mode switches between light and dark display modes. The choice is saved in terminal memory. Check text readability and wallpaper appearance on the device after changing it.

Fonts

1. Open Settings → PIN → **Font**.

2. Each tap selects the next of seven options: **Unicode**, **FreeSans**, **FreeSerif**, **FreeMono**, **FreeSans Bold**, **FreeSerif Bold** and **FreeMono Bold**.

3. Select **Unicode** for languages or employee names containing accented letters. Other fonts may omit these characters.

4. The selected font is automatically saved and restored after restart. The desktop application has no terminal-font selection field.

Language

Tapping **Language** cycles through Polish, English, Spanish and German. The selection is stored in terminal memory. Language can also be set in the SD configuration form.

If an SD language change does not take effect, select the desired language in the terminal menu too: the saved device preference takes priority at startup.

Wi-Fi setup on the terminal

Open Settings → PIN → **Wi-Fi**. Wait for the scan, select a network and enter its password on the touchscreen keyboard. Tap **Connect**. Credentials are written to config.json only after a successful connection; check **Connected; network saved**. In offline mode Wi-Fi turns off again, and the saved network will be used when switching to online. Configure the API URL and token beforehand in the desktop application.

6. Wallpaper: prepare and enable

The desktop application converts a normal image into the terminal format. One wallpaper is used on the home screen and clock-in/out confirmations in both operating modes.

1. Turn off the terminal, connect its SD card to the computer and open the desktop **Wallpaper** tab.

2. Click **Select image**. Choose PNG, JPG, JPEG, BMP or WEBP. A landscape image with a 3:2 aspect ratio gives good control over the crop. The result is **480 × 320 pixels**.

3. Under **Fit mode**, choose **Fill screen / crop** to fill the screen with a centred crop, or **Fit entire image** to retain the whole image with dark background in unused areas.

4. Under **Preview overlays for**, check **Home screen** and **Attendance confirmation**. Toggle the checkboxes beside the preview to inspect areas occupied by interface content.

5. Click **Select SD card folder** in the Wallpaper tab and choose the card root, for example E:\ . This selection is separate from the configuration-tab SD folder.

6. Click **Export wallpaper to SD**. Check the message listing three saved files. Exporting again to the same folder overwrites the previous wallpaper files.

7. Safely eject the card, insert it into the powered-off terminal and restart. The wallpaper is loaded at startup.

8. Open **Settings** → **PIN** → **Wallpaper**. Use the right-hand scrollbar to reach the final items. Tap until Wallpaper shows **ON**. The selection is saved in terminal memory. Check the home screen and a test attendance confirmation.

Exported file	Purpose
WALLPAPER.RGB565	Terminal wallpaper: 480 × 320, exactly 307,200 bytes. Must be in the SD root.
wallpaper_rgb565.h	Additional developer header. Not required to enable wallpaper from SD.
wallpaper_layout.json	Description of fixed interface areas. Does not reposition text or buttons.

Overlays are preview guides only. Selecting screens or checkboxes does not create separate wallpapers or move terminal interface elements. Leave clear space for the clock, date, message, status and employee name. Copying a JPG to SD or renaming it to RGB565 is not sufficient.

If wallpaper does not appear, check WALLPAPER.RGB565, its location in the SD root, the 307,200-byte size, Wallpaper ON and restart after export. Invalid or unreadable files leave the normal background in use. OFF hides the wallpaper without deleting it.

6. Photos, touch and time

Photos at clock-in and clock-out

In the desktop application, select **Enable photos at attendance recording** and save configuration to SD. On the terminal, Settings → PIN → **Photos** toggles ON/OFF and saves the preference in device memory and config.json when SD is available.

ON OK indicates an available camera. **ON !** means photos are enabled but the camera is unavailable. If changing SD configuration does not alter the setting after restart, toggle it in the terminal menu: the stored device preference takes priority.

JPEG images are stored in **photos/** on SD. Names contain the surname, first name, ID, date and time, for example Smith_John_P001_20261011_0800.jpg. An event ID is appended if a name would collide. The photo_file column stores the path. Online mode does not upload photos to Google; a path in a spreadsheet does not share the image. To view photos, connect SD to the computer and open the files in an image viewer.

Camera or photo-save failures are logged; attendance may still be recorded without a photo. A photo may also be created before an unsuccessful online event write, so a photo alone does not confirm attendance. Back up photos and monitor SD free space.

Touch calibration

1. Open Settings → PIN, use the right-hand scrollbar and choose **Calibrate touch**. On first boot, calibration starts automatically if no valid calibration is stored.

2. Touch and hold the centre of each target in order: top-left, top-right, bottom-right, bottom-left. Release before the next target.

3. Allow approximately 30 seconds per point. Successful completion displays **Calibration saved**; the result is stored in terminal memory.

4. Check button accuracy. If calibration fails or times out, repeat the procedure.

Date, time and time-zone offset

In **Date and time**, use + and - beside the hour, minute, day, month and year. Changes adjust the clock immediately. Finish with **Done**.

The SD configuration offset in minutes sets the timestamp UTC offset. For example, Poland normally uses 60 minutes in winter and 120 in summer. There are no automatic daylight-saving rules. When clocks change, check both the terminal clock and configuration offset. Changing the offset alone does not adjust the displayed clock.

7. Records and troubleshooting

Viewing and correcting attendance

Offline: choose the SD root, open **Attendance** and click **Select file from SD**. Select attendance.csv, edit and click **Save / send changes**. Changes are saved to that selected file. In **Logs**, select logs.csv in the same way.

Online: click **Refresh from Google Sheets** in Attendance or Logs. Save attendance corrections using Save / send changes and confirm replacement of the entire table. Always load current data before editing. Logs are read-only.

ENTRY means clock-in; EXIT means clock-out. Preserve existing event_id values; new events need unique IDs. Use ISO 8601 timestamps, for example 2026-10-11T08:00:00+02:00. Use notes to explain a correction. Changing the last event affects the next automatic attendance action.

After directly editing the Google attendance spreadsheet, refresh it in the application and save through the API to rebuild the next-action cache. The application does not provide payroll calculation or a total-hours report.

Common problems

Problem	Action
Online event not confirmed	Check Wi-Fi, API and token. Distinguish queued events from Google confirmation. Wait for synchronization; do not delete pending_online_attendance.csv.
Card not working	Check active, UID and operating mode. Allow online synchronization or restart the terminal.
No door-release option	Check relay_access=true and attendance state. Door release is available after clock-in; online state must be confirmed through the API.
CSV/API header error	Check exact column names and order. Do not translate headers. CSV uses UTF-8 and commas.
Table save rejected	Refresh current data, reapply edits and save. Do not overwrite using an outdated table.
Missing photos / inaccurate touch	Check Photos ON OK, camera, SD and logs. Repeat touch calibration.

Backups and capacity: back up SD or spreadsheets before editing. The terminal loads up to 180 employee records, including inactive ones. Larger tables can leave some cards outside the loaded list. Keep within this limit.

Source basis: the reviewed RCP2027 project; src/main.cpp; desktop/src/rcp2027_app/ui/main_window.py; desktop/src/rcp2027_app/data/schemas.py and wallpaper.py; google_apps_script/Code.gs; README.md and docs/ARCHITECTURE.md.